

What is CSS?

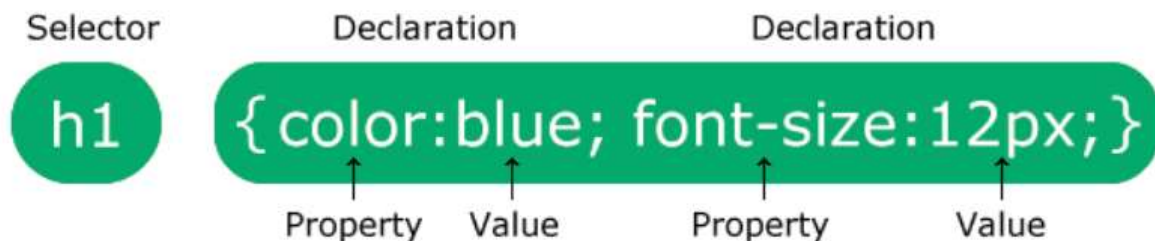
CSS is the language we use to style a Web page.

- CSS stands for Cascading Style Sheets
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files

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Example

In this example all <p> elements will be center-aligned, with a red text color:

```
p {  
  color: red;  
  text-align: center;  
}
```

Example Explained

- **p** is a selector in CSS (it points to the HTML element you want to style: <p>).
- **color** is a property, and **red** is the property value
- **text-align** is a property, and **center** is the property value

You will learn much more about CSS selectors and CSS properties in the next chapter.

CSS Selectors

CSS Selectors are used to **select HTML elements and apply styles** to them. They help the browser identify which elements should receive the specified CSS rules.

The basic syntax of a selector is:

```
selector {  
  property: value;  
}
```

Example:

```
p {  
  color: blue;  
}
```

This selector applies **blue color to all <p> elements**.

1. Element Selector

The **element selector** selects all elements with a specific HTML tag name.

Example:

```
h1 {  
  color: red;  
}
```

HTML:

```
<h1>Hello</h1>  
<h1>Welcome</h1>
```

→ All <h1> elements will appear **red**.

2. Class Selector

The **class selector** selects elements with a specific **class attribute**. It is written using a **dot (.)**.

Example:

```
.highlight {  
    color: green;  
}
```

HTML:

```
<p class="highlight">This is important text</p>
```

→ Only elements with class **highlight** will be styled.

3. ID Selector

The **ID selector** selects a single element with a specific **id attribute**. It is written using **# symbol**.

Example:

```
#title {  
    color: blue;  
}
```

HTML:

```
<h1 id="title">My Website</h1>
```

→ Only the element with **id="title"** will be styled.

4. Universal Selector

The **universal selector (*)** selects **all elements** on the page.

Example:

```
* {  
  margin: 0;  
  padding: 0;  
}
```

→ Applies the style to **every HTML element**.

5. Group Selector

The **group selector** allows applying the same style to **multiple elements**.

Example:

```
h1, h2, p {  
  color: purple;  
}
```

→ All **h1, h2, and p elements** will have purple text.

Selector	Symbol	Example
Element	tag name	p
Class	.	.box
ID	#	#header
Universal	*	*
Group	,	h1, h2

How to Add CSS to HTML

CSS can be added to an HTML document in **three main ways**. These methods control how styles are applied to the webpage.

1. Inline CSS

Inline CSS is written directly inside the HTML element using the **style attribute**.

Example:

```
<p style="color: red; font-size: 20px;">Hello World</p>
```

Explanation:

The style is applied only to that specific element.
Here the paragraph text will appear **red and size 20px**.

Advantages

- Quick for small changes.

Disadvantages

- Not reusable.
 - Makes HTML code messy.
-

2. Internal CSS

Internal CSS is written inside the **<style> tag** in the **<head> section** of the HTML document.

Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: blue;
  font-size: 18px;
}
</style>
</head>
```

```
<body>
<p>This is a paragraph</p>
</body>
</html>
```

Explanation:

All `<p>` elements in the page will have **blue color and 18px font size**.

Advantages

- Styles apply to the whole page.
- Easier to manage than inline CSS.

Disadvantages

- Cannot be reused in multiple pages.
-

3. External CSS

External CSS is written in a **separate .css file** and linked to the HTML document.

Step 1: Create CSS File

Example style.css

```
p {
  color: green;
  font-size: 20px;
}
```

Step 2: Link CSS in HTML

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="style.css">
</head>

<body>
<p>This is a paragraph</p>
</body>
</html>
```

Explanation:

The styles in **style.css** will apply to the HTML page.

Advantages

- Best method for large websites.
- Reusable across many pages.
- Keeps HTML clean.

Short exam answer:

- **rel:** Defines the relationship between the HTML document and the linked file.
- **href:** Specifies the path or URL of the linked resource.

CSS Comments

CSS Comments

Comments are used to explain the CSS code, and may help when you edit the source code at a later date.

Comments are also used to temporarily disable sections of CSS code within a stylesheet.

Comments are ignored by browsers!

A CSS comment is placed inside the HTML `<style>` element, and starts with `/*` and ends with `*/`:

Example

```
/* This is a single-line comment */  
p {  
  color: red;  
}
```

You can add comments wherever you want in the code:

Example

```
p {  
  color: red; /* Set text color to red */  
}
```

Even in the middle of a code line:

Example

```
p {  
  color: /*red*/blue;  
}
```

Comments can also span multiple lines:

Example

```
/* This is  
a multi-line  
comment */
```

```
p {  
  color: red;  
}
```

CSS Colors

CSS Colors are used to **change the color of text, background, borders, and other elements** on a webpage. CSS provides several ways to specify colors.

1. Color Names

CSS supports many **predefined color names**.

Example:

```
h1 {  
  color: red;  
}
```

→ The heading text will appear **red**.

Some common color names:

- red
- blue

- green
- yellow
- black
- white

2. HEX Color

HEX (Hexadecimal) colors use a **6-digit code** starting with #.

Format:

#RRGGBB

Example:

```
p {  
  color: #FF0000;  
}
```

Explanation:

- FF → Red
- 00 → Green
- 00 → Blue

→ This produces **red color**.

Example colors:

- #000000 → Black
- #FFFFFF → White
- #0000FF → Blue

3. RGB Color

RGB stands for **Red, Green, Blue**.

Format:

```
rgb(red, green, blue)
```

Values range from **0 to 255**.

Example:

```
p {  
    color: rgb(255, 0, 0);  
}
```

→ Produces **red color**.

Example:

```
h1 {  
    color: rgb(0, 0, 255);  
}
```

→ Blue color.

CSS Background Color

CSS Background Color is used to **set the background color of an HTML element** such as a page, paragraph, or division.

The CSS property used is:

```
background-color
```

Basic Syntax

```
selector {  
    background-color: color;  
}
```

Example 1: Background Color for Web Page

```
body {  
    background-color: lightblue;  
}
```

→ This sets the **entire webpage background to light blue**

CSS `background-image`

The `background-image` property in CSS is used to **set an image as the background of an HTML element** such as the webpage, a div, or a paragraph.

By default, the background image **repeats (tiles)** both horizontally and vertically to cover the entire element.

Syntax

```
selector {  
  background-image: url("image-name");  
}
```

- **url()** → Specifies the location of the image file.
-

Example

```
body {  
  background-image: url("paper.gif");  
}
```

Explanation:

- The image **paper.gif** will be used as the background of the **entire webpage**.
- The image will **repeat automatically** to fill the page.

Important Related Properties

Property	Purpose
<code>background-image</code>	Sets the background image
<code>background-repeat</code>	Controls image repeating
<code>background-size</code>	Controls image size
<code>background-position</code>	Sets image position

Example:

```
body {  
  background-image: url("paper.gif");  
  background-repeat: no-repeat;  
  background-size: cover;  
}
```